R13

Code No: 115EM

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, May - 2018
SOFTWARE ENGINEERING
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

Used paper 14-05-2018

PART - A

(25 Marks)

1.a)	What is Software Development Life Cycle?	[2]
b)	Mention some of the factors to be considered during System modelling.	[3]
c)	What is meant by Requirement management?	[2]
d)	Differentiate between data flow diagram and state transition diagram.	[3]
e)	List the principles of a software design.	[2]
f)	What are the quality parameters considered for effective modular design?	[3]
g)	What is the role of cyclomatic complexity in software resting?	[2]
h)	Define black box testing strategy?	[3]
i)	Distinguish between reactive and proactive risk management.	[2]
j)	Write short note on RMMM.	[3]

PART - B

(50 Marks)

2.a) What are the advantages of layered technology?
b) Give CMMI levels and explain. [5+5]

OR

3.a) How does system engineering differ from software engineering? Also write brief notes on computer based system and system engineering hierarchy.
b) Explain in detail Evolutionary process model. [5+5]

4.a) Why is traceability an important aspect of requirement management? Why context system models are useful for requirements validation?
b) Explain about the cardinality and modality with suitable example. [5+5]

OR

5. Give an overview of various steps in requirements engineering process. [10]

6.a) Write about architectural styles and patterns.
b) Explain interface analysis and interface design steps. [5+5]

OR

7.a) How a component is designed based on function ? Explain.
b) What are the golden rules for user interface design? Explain. [5+5]

8.a) What are the strategic approaches to software testing?
b) Discuss about art of debugging in detail. [5+5]

OR

9.a) What is software testing? Explain clearly the system testing.
b) What is meant by software quality? Explain the metrics for maintenance. [5+5]

10.a) How risk is identified? Explain.
b) Discuss about software reviews. [5+5]

OR

11.a) Discuss about Formal technical reviews.
b) What are the types of software risks? [5+5]

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